

Companion to Bird & de Moor, Algebra of Programming

§6. Recursive Programs

§6.1. Digits of a number

(6.1a)

$val^\circ = embed^\circ wrap \cup op^\circ -xDigit(val^\circ)snoc$

a number is read back into digits either as one nonzero digit, or by splitting off its last digit and reading back the rest

(a)

(b)

(c)

(d)

(e)

(a)	val°	val undoes $wrap$ or $snoc$, reads the front with val and combines by $embed$ or op , so its converse runs that backwards
(b)	$([wrap, snoc]^\circ Digit^{+-} (-xDigit(val)) [embed, op])^\circ$	undoing $wrap$ or $snoc$ first gives two cases: a decimal and its last digit read by val then op , or one digit read by $embed$
(c)	$(wrap^\circ embed)^\circ \cup (snoc^\circ -xDigit(val) op)^\circ$	the converse of each case is its steps undone in reverse: undo op , undo reading the front, then $snoc$; or undo $embed$, then $wrap$
(d)	$embed^\circ wrap \cup op^\circ -xDigit(val)^\circ snoc$	undoing the read of the front while keeping the last digit is val° on the front alone
(e)	$embed^\circ wrap \cup op^\circ -xDigit(val^\circ) snoc$	

(6.1b)

$op^\circ(m, p) \iff \pi_1(p) = m/10 \wedge \pi_2(p) = m \% 10$
 op° splits a number into its quotient and remainder by 10
 $(\exists p. 0 < \pi_1 p \wedge op^\circ(m, p)) \iff 10 \leq m$
 op° gives a pair with a nonzero first component exactly at the numbers with two or more digits
 $(\exists d. embed^\circ(m, d)) \iff 0 < m \wedge m < 10$
 $embed^\circ$ gives a digit exactly at the one-digit numbers

§6.2. Least fixed points

(6.2a)

φ a **monotonic** mapping of the hom-set $A \rightarrow B$ into itself: $X \sqsubseteq Y \Rightarrow \varphi(X) \sqsubseteq \varphi(Y)$.

$(\mu X : \varphi(X))$ the least $X : A \rightarrow B$ with $\varphi(X) \sqsubseteq X$.

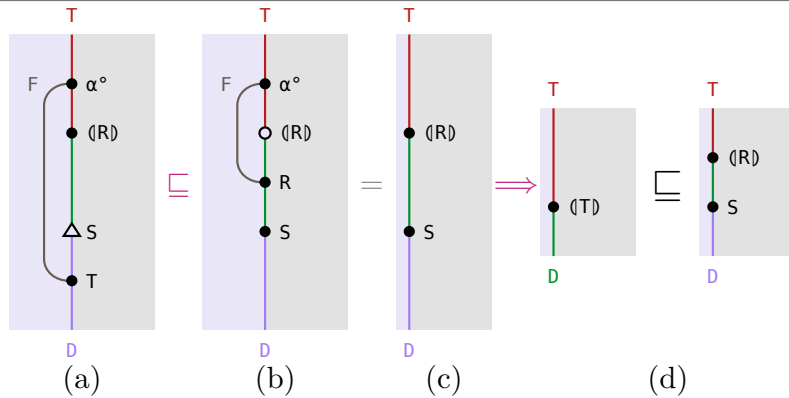
(6.2b)

the law	what it says
$\varphi(Y) \sqsubseteq Y \Rightarrow (\mu X : \varphi(X)) \sqsubseteq Y$	to bound $(\mu X : \varphi(X))$ above, exhibit one Y the body does not grow past — the half §Section 6.3 and every chapter after it uses
$\text{Monotonic}(\varphi) \Rightarrow \varphi(\mu X : \varphi(X)) = (\mu X : \varphi(X))$	Knaster–Tarski : the least solution of $\varphi(X) \sqsubseteq X$ already solves $\varphi(X) = X$, so the least prefix point and the least fixed point are one relation
$\alpha^\circ F(X) R \sqsubseteq X \Rightarrow \langle R \rangle \sqsubseteq X$ (6.2)	a relation X that contains one unfolding of the fold's recursion at X contains the fold
$X \sqsubseteq \alpha^\circ F(X) R \Rightarrow X \sqsubseteq \langle R \rangle$ (6.3)	a relation X contained in one unfolding of the fold's recursion at X is contained in the fold

$$F(S)T \sqsubseteq RS \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$$

(6.2c)

if S followed by R absorbs T after $F(S)$, the fold of T is below the fold of R followed by S

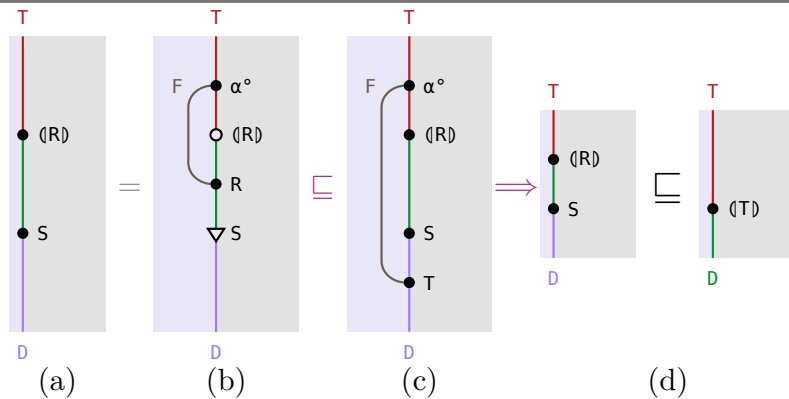


(a)	$\alpha^\circ F(\langle R \rangle) F(S) T$	$F(S) T \sqsubseteq RS$
(b)	$\alpha^\circ F(\langle R \rangle) RS$	$F(\langle R \rangle) R = \alpha(\langle R \rangle), \alpha^\circ \alpha = 1 \triangleq \text{reduce}$
(c)	$\langle R \rangle S$	$\langle R \rangle S$ is a prefix point of $\alpha^\circ F(-) T$, and $\langle T \rangle$ is the least one — (6.2) $\alpha^\circ F(X) T \sqsubseteq X \Rightarrow \langle T \rangle \sqsubseteq X$
(d)	$\alpha^\circ F(\langle R \rangle S) T \sqsubseteq \langle R \rangle S \Rightarrow \langle T \rangle \sqsubseteq \langle R \rangle S$	

$$R \sqsubseteq F(S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$$

(6.2d)

if R followed by S is absorbed by $F(S)$ followed by T , the fold of R followed by S is below the fold of T



(a)	$\langle R \rangle S$	$\alpha(\langle R \rangle) = F(\langle R \rangle) R, \alpha^\circ \alpha = 1 \triangleq \text{reduce}$
(b)	$\alpha^\circ F(\langle R \rangle) RS$	$R \sqsubseteq F(S) T$
(c)	$\alpha^\circ F(\langle R \rangle S) T$	$\langle R \rangle S$ is a postfix point of $\alpha^\circ F(-) T$, and $\langle T \rangle$ is the greatest one — (6.3) $X \sqsubseteq \alpha^\circ F(X) T \Rightarrow X \sqsubseteq \langle T \rangle$
(d)	$\langle R \rangle S \sqsubseteq \alpha^\circ F(\langle R \rangle S) T \Rightarrow \langle R \rangle S \sqsubseteq \langle T \rangle$	

§6.3. Hylomorphisms

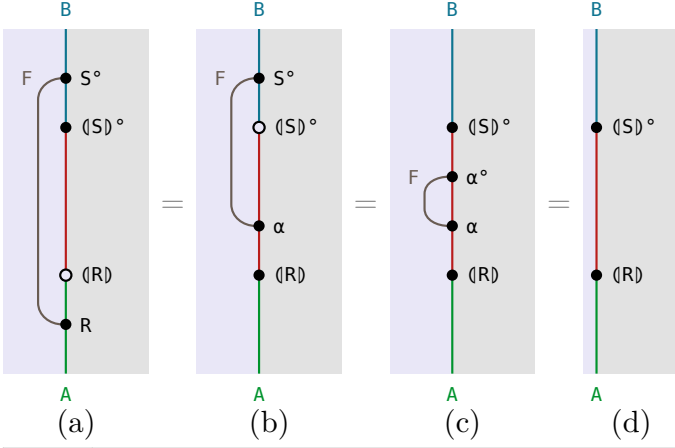
$$(\mathbb{S})^\circ(\mathbb{R}) = (\mu X : S^\circ F(X)R)$$

(6.3a)

hylomorphism theorem: a hylomorphism is the least fixed point of a certain recursion equation

$$S^\circ F((\mathbb{S})^\circ(\mathbb{R}))R = (\mathbb{S})^\circ(\mathbb{R})$$

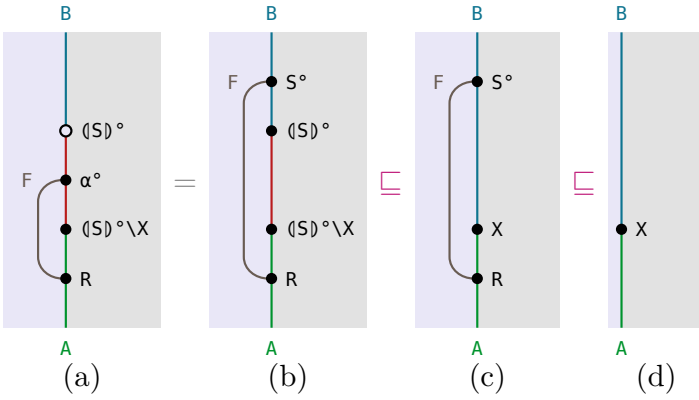
hylomorphism theorem: a prototypical ‘divide and conquer’ scheme — the term S° represents the decomposition stage, $F((\mathbb{S})^\circ(\mathbb{R}))$ the stage of solving the subproblems recursively, and R the recombination stage; $R : FA \rightarrow A$, $S : FB \rightarrow B$, $\alpha : FT \rightarrow T$ initial



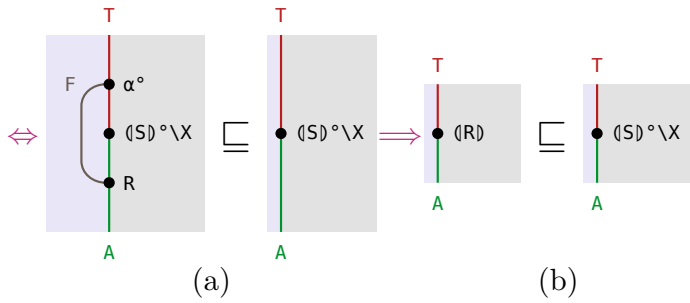
(a)	$S^\circ F((\mathbb{S})^\circ)F((\mathbb{R}))R$	$F((\mathbb{R}))R = \alpha((\mathbb{R})) \triangleq \text{reduce}$
(b)	$S^\circ F((\mathbb{S})^\circ)\alpha((\mathbb{R}))$	$(\mathbb{S})^\circ\alpha^\circ = S^\circ F((\mathbb{S})^\circ) \triangleq \text{reduce, relator-laws}$
(c)	$(\mathbb{S})^\circ\alpha^\circ\alpha((\mathbb{R}))$	$\alpha \text{ iso}$
(d)	$(\mathbb{S})^\circ(\mathbb{R})$	

$$S^\circ F(X)R \sqsubseteq X \implies (\mathbb{S})^\circ(\mathbb{R}) \sqsubseteq X$$

hylomorphism theorem: by Knaster–Tarski, the hylomorphism $(\mathbb{S})^\circ(\mathbb{R})$ is included in X if X satisfies the associated recursion inequation



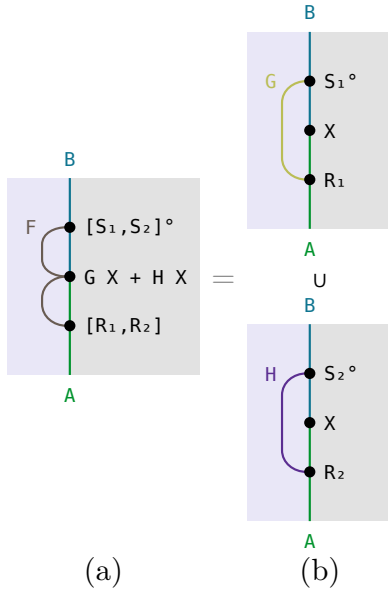
(a)	$(\mathbb{S})^\circ\alpha^\circ F((\mathbb{S})^\circ\backslash X)R$	$(\mathbb{S})^\circ\alpha^\circ = S^\circ F((\mathbb{S})^\circ)$
(b)	$S^\circ F((\mathbb{S})^\circ((\mathbb{S})^\circ\backslash X))R$	$(\mathbb{S})^\circ((\mathbb{S})^\circ\backslash X) \sqsubseteq X \triangleq \text{adj-all}$
(c)	$S^\circ F(X)R$	$S^\circ F(X)R \sqsubseteq X$
(d)	X	$S \cdot \dashv S \triangleq \text{adj-all}$



(a)	$S^\circ F(X)R \sqsubseteq X \implies \alpha^\circ F((\mathbb{S})^\circ\backslash X)R \sqsubseteq (\mathbb{S})^\circ\backslash X$	$(\mathbb{R}) = (\mu X : \alpha^\circ F(X)R) \triangleq \text{reduce, (6.2b), adj-all}$
(b)	$S^\circ F(X)R \sqsubseteq X \implies (\mathbb{R}) \sqsubseteq (\mathbb{S})^\circ\backslash X$	

$$(\forall x. \forall y. \forall X. F(X)=G(X)+H(X)) \Rightarrow ([S_1, S_2]^\circ \circ ([R_1, R_2])^\circ) = (\mu X : S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2) \quad (6.3b)$$

when both algebras are case splits over the same sum, the recursion runs each case on its own and unites the results

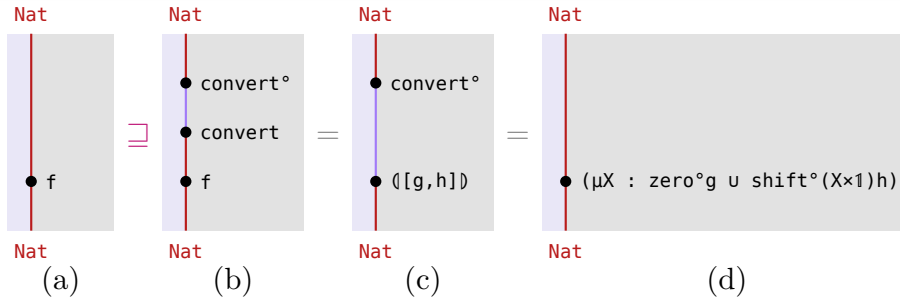


(a)	$[S_1, S_2]^\circ (G(X) + H(X)) [R_1, R_2]$	$[S, T] = (S + T) \nabla$ on both sides gives $\nabla^\circ (S_1^\circ G(X) R_1 + S_2^\circ H(X) R_2) \nabla$; the + stands for the codiagonals ∇°, ∇ , which are not drawn
(b)	$S_1^\circ G(X) R_1 \cup S_2^\circ H(X) R_2$	$\nabla^\circ (P + Q) \nabla = P \cup Q$ — coproduct

§6.4. Fast exponentiation and modulus computation

$$\text{zero } f = g \wedge \text{shift } f = (f \times 1)h \Rightarrow (\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1)h) \sqsubseteq f \quad (6.4a)$$

once **f** sends **zero** to **g** and turns **shift** into **h**, the recursion that halves the argument at each step computes **f**



(a)	f	convert simple
(b)	$\text{convert}^\circ \text{convert } f$	fusion: $\text{zero } f = g, \text{shift } f = (f \times 1)h$
(c)	$\text{convert}^\circ ([g, h])$	Corollary 6.1, (6.3b)
(d)	$(\mu X : \text{zero}^\circ g \cup \text{shift}^\circ (X \times 1)h)$	

$$(\mu X : \text{zero}^\circ \text{one} \cup \text{shift}^\circ (X \times 1) \text{op}(a)) \sqsubseteq \text{exp}(a) \quad (6.4b)$$

the recursion that halves the exponent at each step computes **a** to the power **b**

$\text{zero } \text{exp}(a) = \text{one}$

a to the power 0 is 1

$\text{shift } \text{exp}(a) = (\text{exp}(a) \times 1) \text{op}(a)$

a to the power $2n+d$ is $\text{op}(a)$ of a to the power n and d

$$(\mu X : \text{zero}^\circ \text{zero} \cup \text{shift}^\circ (X \times 1) \text{op}(b)) \sqsubseteq \text{mod}(b) \quad (6.4c)$$

the recursion that halves **a** at each step computes the remainder of **a** by **b**

$\text{zero } \text{mod}(b) = \text{zero}$

$0 \bmod b$ is 0

$\text{shift } \text{mod}(b) = (\text{mod}(b) \times 1) \text{op}(b)$

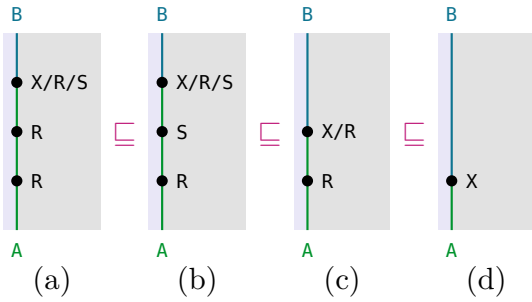
$(2a+d) \bmod b$ is $\text{op}(b)$ of $a \bmod b$ and d

§6.5. Unique fixed points

$$\text{inductive}(S) \wedge RR \sqsubseteq SR \Rightarrow \text{inductive}(R)$$

(6.5a)

R is inductive when $RR \sqsubseteq SR$ for an inductive S



(a)	$(X/R/S)RR$	$RR \sqsubseteq SR$
(b)	$(X/R/S)SR$	division by S
(c)	$(X/R)R$	division by R
(d)	X	

$$\text{member} = \emptyset$$

(6.5b)

a constant relator records no elements, so its membership is empty

$$\text{member} = [\text{member}, \text{member}]$$

a member of $F+G$ is a member of whichever summand the value is in

$$\text{member} = \pi_1 \text{ member} \cup \pi_2 \text{ member}$$

a member of $F \times G$ is a member of either component

$$\text{inductive}(S \text{ member}) \wedge X = SF(X)R \wedge Y = SF(Y)R \Rightarrow X = Y$$

(6.5c)

when $S \text{ member}$ is inductive, two solutions X, Y of $X = SF(X)R$ are equal

$$\text{inductive}(S \text{ member}) \wedge \text{Entire}(S) \wedge \text{Entire}(R) \wedge SF(X)R \in X \Rightarrow \text{Entire}(X)$$

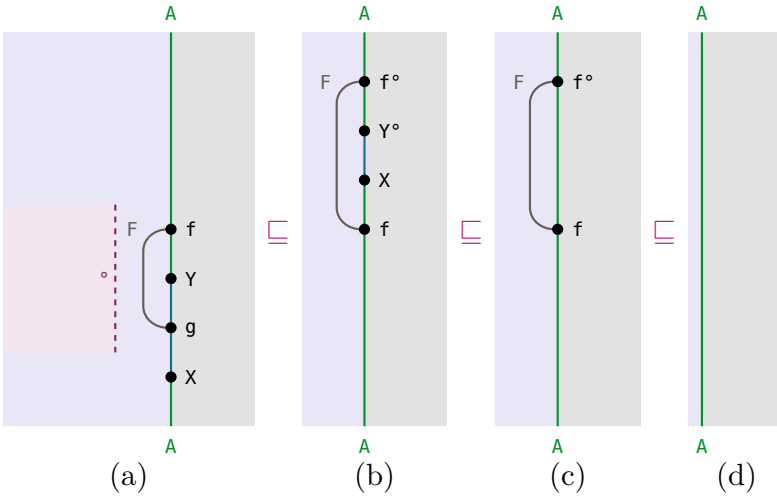
(6.5d)

when $S \text{ member}$ is inductive and S, R are entire, a pre-fixed point of $X \mapsto SF(X)R$ is entire

$$\text{inductive}(g \text{ member}) \wedge \text{Map}(g) \wedge \text{Map}(f) \wedge X = gF(X)f \Rightarrow \text{Map}(X)$$

(6.5e)

when $g \text{ member}$ is inductive and f, g are maps, the solution of $X = gF(X)f$ is a map

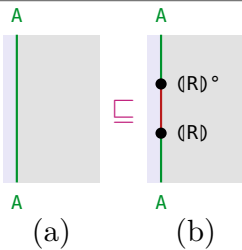


(a)	$(gF(Y)f) \circ X$	$X = gF(X)f, g \text{ simple}$
(b)	$f \circ F(Y \circ X)f$	$Y \circ X \in \mathbb{I}$
(c)	$f \circ f$	$f \text{ simple}$
(d)	1	

$$\text{inductive}(R^\circ \text{ member}) \wedge 1 \in R^\circ R \Rightarrow 1 \in (R)^\circ (R)$$

(6.5f)

when $R^\circ \text{ member}$ is inductive and R is surjective, the fold (R) is surjective

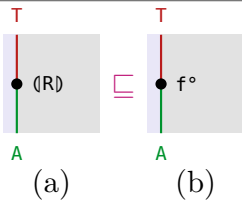


(a)	1	(6.5d) at $S = R^\circ$
(b)	$(R)^\circ (R)$	

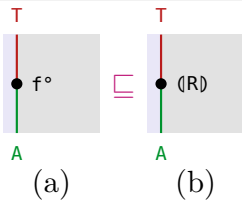
$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \wedge 1 \sqsubseteq R^\circ R \Rightarrow f^\circ = \langle R \rangle$$

(6.5g)

a map f with $Rf \sqsubseteq F(f)\alpha$, for a surjective R , has the fold $\langle R \rangle$ as its converse



(a)	$\langle R \rangle$	shunting f , fusion, $Rf \sqsubseteq F(f)\alpha$
(b)	f°	

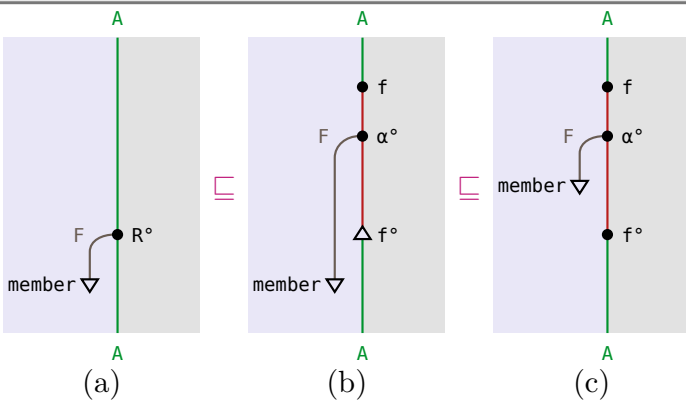


(a)	f°	(6.5f) and $\langle R \rangle f \sqsubseteq 1$
(b)	$\langle R \rangle$	

$$\text{Map}(f) \wedge Rf \sqsubseteq F(f)\alpha \Rightarrow R^\circ \text{member} \sqsubseteq f\alpha^\circ \text{member } f^\circ$$

(6.5h)

$R^\circ \text{member}$ is below $\alpha^\circ \text{member}$ conjugated by f , so it is inductive when $\alpha^\circ \text{member}$ is



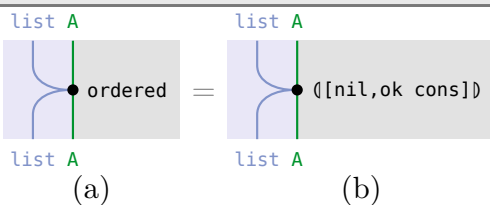
(a)	$R^\circ \text{member}$	$Rf \sqsubseteq F(f)\alpha$, shunting
(b)	$f\alpha^\circ F(f^\circ) \text{member}$	member lax natural
(c)	$f\alpha^\circ \text{member } f^\circ$	

§6.6. Sorting by selection

$$\text{ordered} = \langle [\text{nil}, \text{ok cons}] \rangle$$

(6.6a)

a list is sorted exactly when the fold that rebuilds it passes **ok** at every **cons**, i.e. each head is **R**-below every element after it

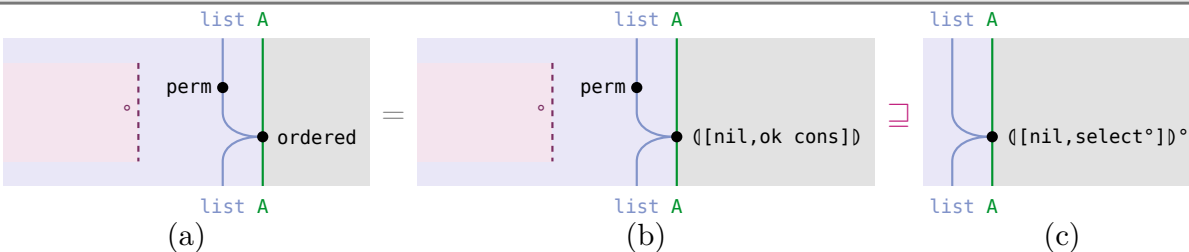


(a)	ordered	fold uniqueness
(b)	$\langle [\text{nil}, \text{ok cons}] \rangle$	

$$\text{select}^\circ \sqsubseteq \text{ok cons perm} \Rightarrow \langle [\text{nil}, \text{select}^\circ] \rangle^\circ \sqsubseteq \text{perm ordered}$$

(6.6b)

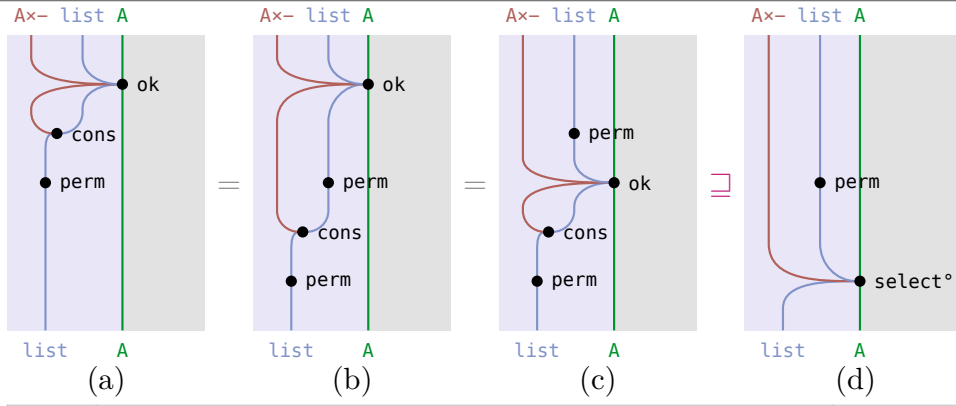
every output of unfolding the input by **select** is a sorted permutation of it



(a)	$(\text{ordered perm})^\circ$	(6.6a)
(b)	$(\langle [\text{nil}, \text{ok cons}] \rangle \text{perm})^\circ$	fusion (6.4) under (6.6c)
(c)	$\langle [\text{nil}, \text{select}^\circ] \rangle^\circ$	

$$\text{select}^\circ \text{ok cons perm} \Rightarrow (1 \times \text{perm}) \text{select}^\circ \text{ok cons perm} \quad (6.6c)$$

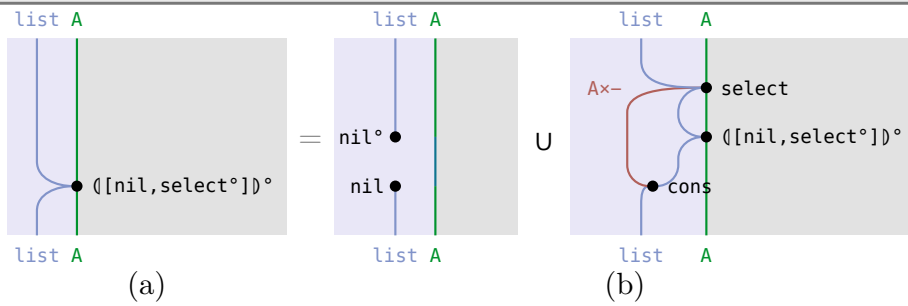
permuting the tail and then undoing **select** lands among the **ok** conses of a permutation



(a)	ok cons perm	cons perm = (1 × perm) cons perm
(b)	ok (1 × perm) cons perm	Ex 6.22, ok (1 × perm) = (1 × perm) ok
(c)	(1 × perm) ok cons perm	select [°] ok cons perm
(d)	(1 × perm) select [°]	

$$([nil, \text{select}^\circ])^\circ = nil^\circ nil \cup \text{select} (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons} \quad (6.6d)$$

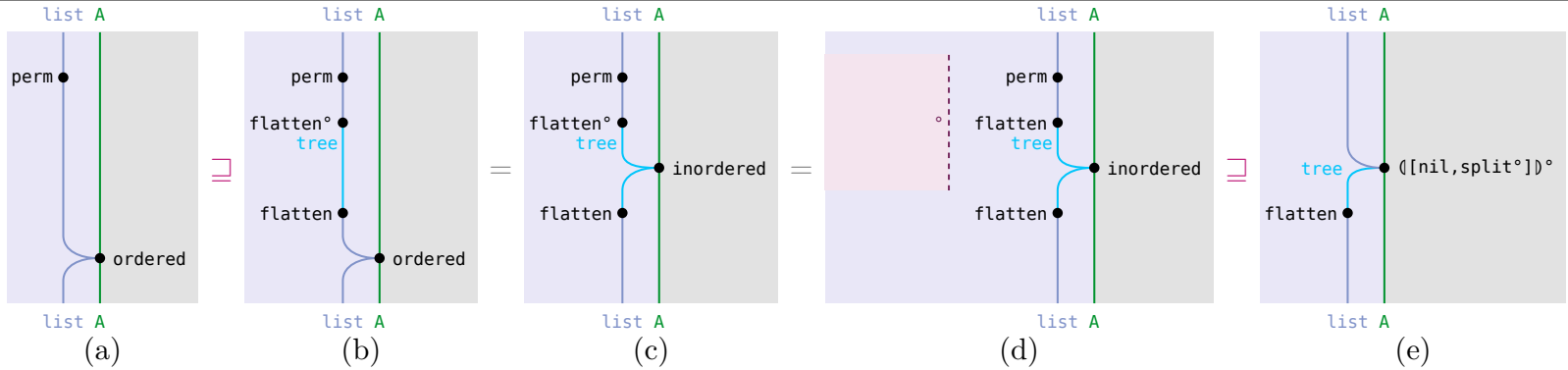
selection sort returns [] on [], and otherwise selects (a,y), sorts y and conses a back on



(a)	$([nil, \text{select}^\circ])^\circ$	unfolding the converse of a fold
(b)	$nil^\circ nil \cup \text{select} (1 \times ([nil, \text{select}^\circ])^\circ) \text{cons}$	

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \wedge \text{split}^\circ \text{check}' \text{ join perm} \Rightarrow ([nil, \text{split}^\circ])^\circ \text{flatten} \text{Eperm ordered} \quad (6.6e)$$

every output of unfolding the input by **split** into a tree and flattening that tree is a sorted permutation of the input

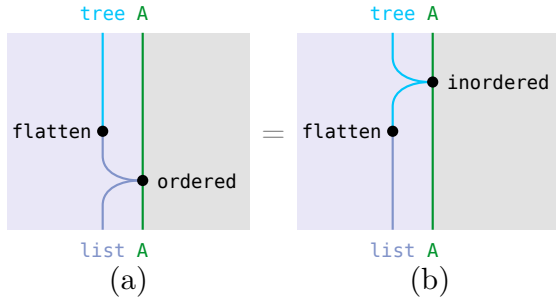


(a)	perm ordered	flatten is simple
(b)	perm flatten [°] flatten ordered	flatten ordered = inordered flatten
(c)	perm flatten [°] inordered flatten	converses
(d)	(inordered flatten perm) [°] flatten	fusion (6.4) under (6.6j)
(e)	$([nil, \text{split}^\circ])^\circ \text{flatten}$	

$$(\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow \text{flatten ordered} = \text{inordered flatten}$$

(6.6f)

flattening a tree and then testing the list for order is the same as testing the tree with **inordered** and then flattening

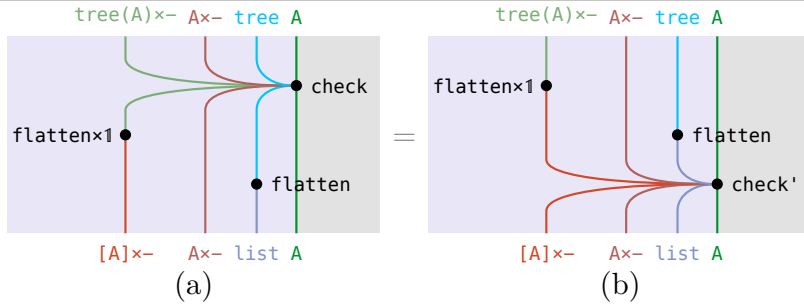


(a)	flatten ordered	R transitive: a flattened tree is ordered iff every node passes check
(b)	inordered flatten	

$$\text{check (flatten} \times (1 \times \text{flatten))} = (\text{flatten} \times (1 \times \text{flatten})) \text{check'}$$

(6.6g)

testing a fork with **check** and then flattening both subtrees equals flattening both subtrees and then testing with **check'**, since **intree** of a tree is **inlist** of its flattening

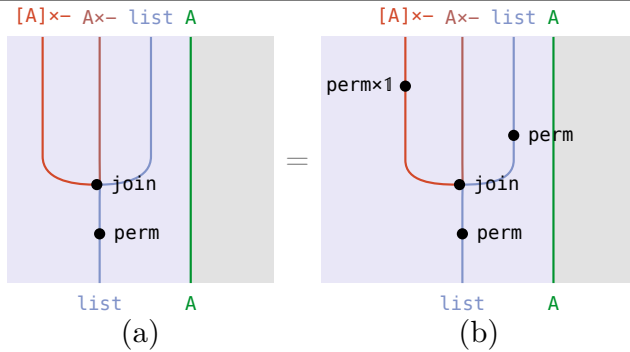


(a)	check (flatten $\times$ (1 $\times$ flatten))	b intree x $\iff$ b inlist flatten(x)
(b)	(flatten $\times$ (1 $\times$ flatten)) check'	

$$\text{join perm} = (\text{perm} \times (1 \times \text{perm})) \text{join perm}$$

(6.6h)

permuting the list **x ++ [a] ++ y** is the same as first permuting **x** and **y** and then permuting the joined list

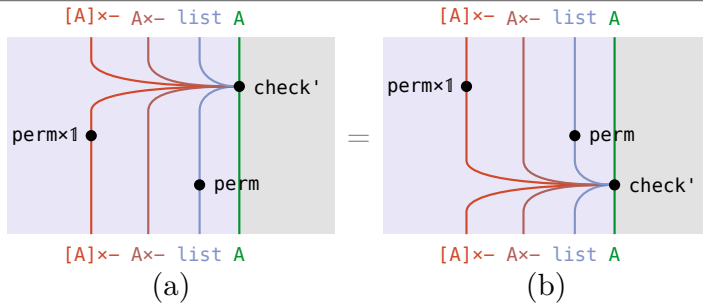


(a)	join perm	perm is transitive and closed under ++
(b)	(perm $\times$ (1 $\times$ perm)) join perm	

$$\text{check' (perm} \times (1 \times \text{perm))} = (\text{perm} \times (1 \times \text{perm})) \text{check'}$$

(6.6i)

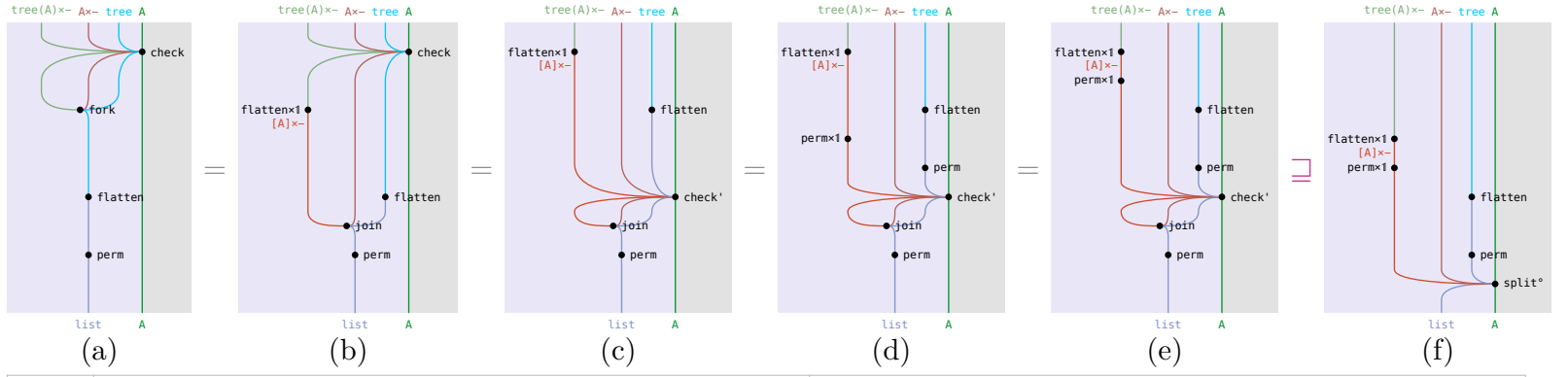
testing a fork of lists with **check'** and then permuting both parts equals permuting both parts and then testing, since permuting a list leaves its members unchanged



(a)	check' (perm $\times$ (1 $\times$ perm))	b inlist x $\iff$ b inlist perm(x)
(b)	(perm $\times$ (1 $\times$ perm)) check'	

split^ocheck' join perm $\Rightarrow ((\text{flatten perm}) \times (1 \times (\text{flatten perm}))) \text{split}^o \text{check fork flatten perm}$ (6.6j)

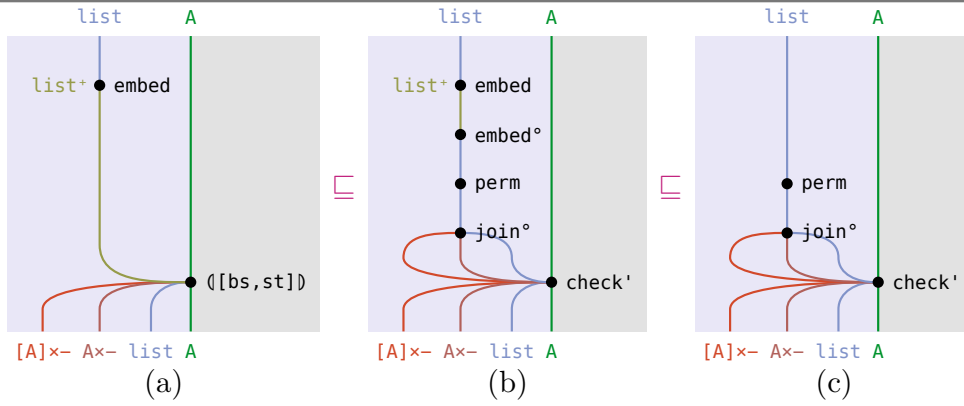
undoing **split** and then flattening and permuting both parts lands among the **checked** forks whose flattening is permuted



(a)	check fork flatten perm	fork flatten = F(flatten)join
(b)	check (flatten × (1 × flatten))join perm	check F(flatten) = F(flatten)check'
(c)	(flatten × (1 × flatten))check' join perm	join perm = F(perm)join perm, check' F(perm) = F(perm)check'
(d)	(flatten × (1 × flatten))(perm × (1 × perm))check' join perm	functors
(e)	((flatten perm) × (1 × (flatten perm)))check' join perm	split ^o check' join perm
(f)	((flatten perm) × (1 × (flatten perm)))split ^o	

bs_≡wrap perm join^ocheck' ∧ (1 × (perm join^ocheck'))st_≡cons perm join^ocheck' $\Rightarrow \text{embed}([bs, st])_{\equiv}$ (6.6k)

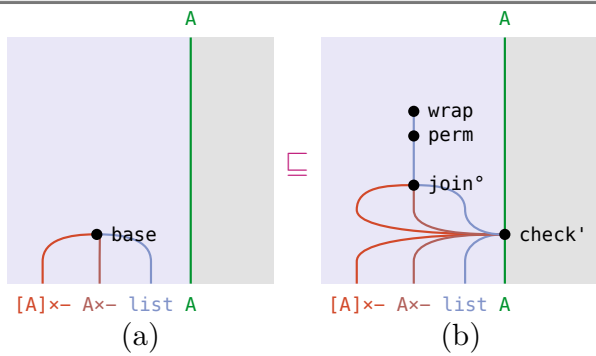
turning a list into a non-empty list and folding it with any **[bs, st]** meeting the two conditions below gives only **checked triples (x, a, y)** whose join $x \# [a] \# y$ is a permutation of the list



(a)	embed([bs, st])	(6.6l) and (6.6m), induction on the non-empty list
(b)	embed embed ^o perm join ^o check'	embed is simple
(c)	perm join ^o check'	

base_≡wrap perm join^ocheck' (6.6l)

base(a) is a **checked triple** whose join is a permutation of the one-element list **[a]**

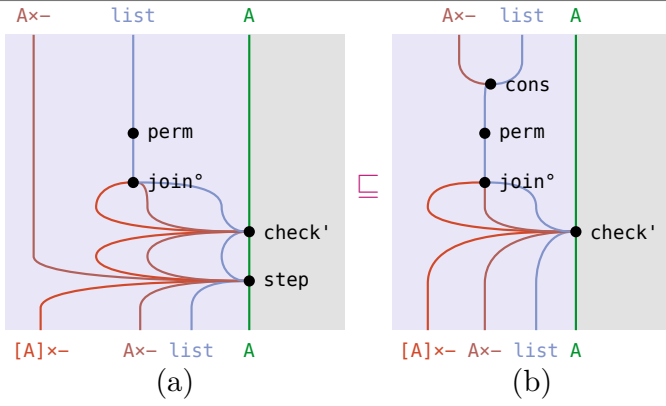


(a)	base	$[] \# [a] \# [] = [a]$, and check' holds on empty lists
(b)	wrap perm join ^o check'	

(6.6m)

$(\forall a. \forall b. \text{leb}(a,b)=\text{true} \Rightarrow R(a,b)) \wedge (\forall a. \forall b. \text{leb}(a,b)=\text{false} \Rightarrow R(b,a)) \Rightarrow (1 \times (\text{perm} \text{ join}^\circ \text{ check}')) \text{ step} \sqsubseteq \text{cons perm join}^\circ \text{ check}'$

if (x,b,y) is a **checked** triple for a permutation of $\mathbf{l}$, then $\text{step}(a, (x,b,y))$ is a **checked** triple for a permutation of $[a] \# \mathbf{l}$



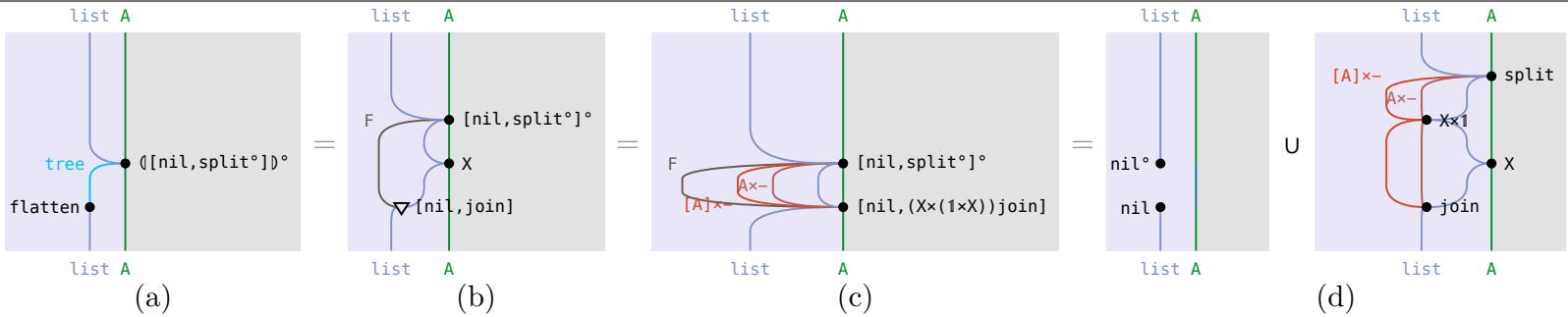
(a)	$(1 \times (\text{perm join}^\circ \text{ check}')) \text{ step}$	aRb puts a in front of x ; otherwise bRa and a goes in front of y
(b)	$\text{cons perm join}^\circ \text{ check}'$	

(6.6n)

$(\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{ flatten} = \text{nil}^\circ \text{ nil} \cup \text{split}$

$((\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{ flatten}) \times (1 \times (\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{ flatten})) \text{ join}$

quicksort returns $[]$ on $[]$, and otherwise splits into (x,a,y) , sorts x and y and joins them around a

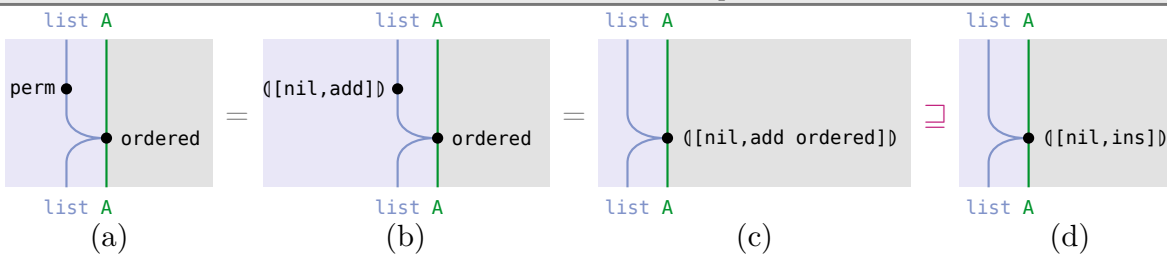


(a)	$(\llbracket \text{nil}, \text{split}^\circ \rrbracket)^\circ \text{ flatten}$	hylomorphism theorem
(b)	$\llbracket \text{nil}, \text{split}^\circ \rrbracket^\circ F(X) \llbracket \text{nil}, \text{join} \rrbracket$	$F(X) \llbracket \text{nil}, \text{join} \rrbracket = \llbracket \text{nil}, (X \times 1 \times X) \text{ join} \rrbracket$
(c)	$\llbracket \text{nil}, \text{split}^\circ \rrbracket^\circ \llbracket \text{nil}, (X \times (1 \times X)) \text{ join} \rrbracket$	coproduct
(d)	$\text{nil}^\circ \text{ nil} \cup \text{split } (X \times (1 \times X)) \text{ join}$	

(6.6o)

$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered} \Rightarrow (\llbracket \text{nil}, \text{ins} \rrbracket) \sqsubseteq \text{perm ordered}$

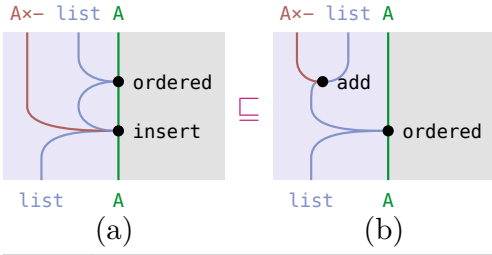
for any **ins** that, on an ordered list, returns only ordered results of **add**, folding with $[\text{nil}, \text{ins}]$ gives only sorted permutations



(a)	perm ordered	$\text{perm} = (\llbracket \text{nil}, \text{add} \rrbracket)$
(b)	$(\llbracket \text{nil}, \text{add} \rrbracket) \text{ ordered}$	fusion, $\text{add ordered} = (1 \times \text{ordered}) \text{ add ordered}$
(c)	$(\llbracket \text{nil}, \text{add ordered} \rrbracket)$	$(1 \times \text{ordered}) \text{ ins} \sqsubseteq \text{add ordered}$
(d)	$(\llbracket \text{nil}, \text{ins} \rrbracket)$	

$$(\forall a. \forall b. \text{leb}(a,b)=\text{true} \Rightarrow R(a,b)) \wedge (\forall a. \forall b. \text{leb}(a,b)=\text{false} \Rightarrow R(b,a)) \wedge (\forall a. \forall b. \forall c. R(a,b) \Rightarrow R(b,c) \Rightarrow R(a,c)) \Rightarrow (1 \times \text{ordered}) \text{insert} \sqsubseteq \text{add ordered} \quad (6.6p)$$

inserting **a** into an ordered list puts **a** somewhere in it and keeps it ordered



(a)	$(1 \times \text{ordered}) \text{insert}$	insert splices a in, and R transitive and total
(b)	add ordered	

§6.7. Closure

$$R^* = (\mu X : 1 \cup XR) \quad (6.7a)$$

closing **R** by composing it on the left and closing it by composing on the right give the same relation **R***

$$1 \in X \wedge XX \in X \Rightarrow REX \Leftrightarrow R^*EX \quad (6.7b)$$

for a preorder **X**, **X** contains **R** exactly when it contains **R***

$$1 \in R^* \quad (6.7c)$$

R* is reflexive

	(a)	1	
$\sqsubseteq$	(b)	$1 \cup RR^*$	union
$=$	(c)	R^*	fixed point, Thm 6.1

$$R \in R^* \quad (6.7d)$$

R* contains **R**

	(a)	R	
$=$	(b)	R	identity
$\sqsubseteq$	(c)	RR^*	$1 \in R^* \text{ — (6.7c)}$
$\sqsubseteq$	(d)	R^*	fixed point

$$R^*R^* \in R^* \quad (6.7e)$$

R* is transitive, because **R*/R*** is a prefixed point of $X \mapsto 1 \cup RX$

	(a)	$(1 \cup R(R^*/R^*))R^*$	
$=$	(b)	$R^* \cup R(R^*/R^*)R^*$	$(P \cup Q)T = PT \cup QT \text{ — composition distributes over } \cup$
$\sqsubseteq$	(c)	$R^* \cup RR^*$	$(S/S)S \in S$
$\sqsubseteq$	(d)	R^*	$RR^* \in R^* \text{ — (6.7d)}$

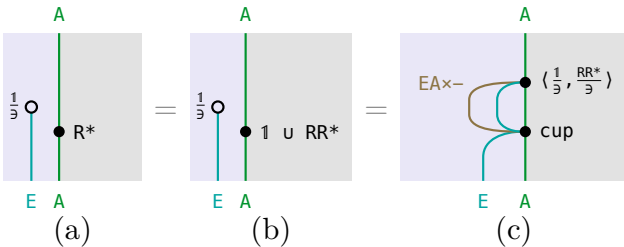
$$1 \in X \wedge XX \in X \wedge REX \Rightarrow R^*EX \quad (6.7f)$$

a reflexive transitive **X** containing **R** is a prefixed point of $X \mapsto 1 \cup RX$, so it contains **R***

	(a)	$1 \cup RX$	
$\sqsubseteq$	(b)	$X \cup XX$	$1 \in X, R \in X$
$\sqsubseteq$	(c)	X	$XX \in X$

$$\frac{R^*}{3} = \left(\frac{1}{3}, \frac{RR^*}{3} \right) \text{cup} \quad (6.7g)$$

the set of **R***-successors of **a** is **a** itself joined with the **R***-successors of its **R**-successors



(a)	$\frac{R^*}{3}$	$R^* = 1 \cup RR^*$
(b)	$\frac{1 \cup RR^*}{3}$	$\wedge(R \cup S) = (\wedge(R), \wedge(S)) \text{ cup}$
(c)	$\left(\frac{1}{3}, \frac{RR^*}{3} \right) \text{cup}$	

$R - \emptyset = R$	(6.7h)
taking nothing away leaves R	
$R \cup (S - R) = R \cup S$	
beside R , only the part of S outside R adds anything	
$R - (S \cup T) = R - S - T$	
taking away S and then T takes away $S \cup T$	
$(R \cup S) - T = (R - T) \cup (S - T)$	
taking T away from a union takes it away from each part	

$\text{monotonic}(\varphi) \wedge \text{monotonic}(\psi) \implies (\mu X : \varphi(\psi(X))) = \varphi(\mu Y : \psi(\varphi(Y)))$	(6.7i)
the least fixed point of φ after ψ is φ applied to the least fixed point of ψ after φ	

$SR^*=(\mu X : S \cup XR)$	(6.7j)
S followed by any number of R steps is the least X containing S and closed under a further R step	
$R^*S=(\mu X : S \cup RX)$	
any number of R steps followed by S is the least X containing S and closed under an R step in front	

$\theta(\emptyset, S) = SR^*$ (6.7k)			
started with nothing found and S to explore, θ computes SR^*			
	(a)	$\theta(\emptyset, S)$	
=	(b)	$\emptyset \cup (\mu X : S \cup (XR - \emptyset))$	definition of θ
=	(c)	$(\mu X : S \cup XR)$	$R - \emptyset = R$ — (6.7h)
=	(d)	SR^*	(6.7j)

$\theta(P, \emptyset) = P$ (6.7l)			
with nothing left to explore, θ returns what it has found			
	(a)	$\theta(P, \emptyset)$	
=	(b)	$P \cup (\mu X : \emptyset \cup (XR - P))$	definition of θ
=	(c)	$P \cup \emptyset$	$\emptyset$ is a prefixed point: $\emptyset R - P = \emptyset$
=	(d)	P	$P \cup \emptyset = P$